

## GENERAL THEORY OF COHERENT EXCITATIONS ON BIOLOGICAL SYSTEMS

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### ABSTRACT

Coherent excitations, as first proposed in 1968, can arise in the presence of nonlinear interactions if metabolic energy is supplied at a rate  $s$  above a critical  $s_0$ . Three types of excitations will be shown to exist: A excitation of a single mode; B excitation of a metastable highly polar state; C excitation of limit cycles or Lotka-Volterra oscillations in complex systems. They may have far reaching consequences on the dynamics of biological systems.

Externally applied radiation may, at certain frequencies and at certain stages of development, act as trigger to biological events. In this case a steplike dependence on the intensity is to be expected.

Relevant experimental evidence will be indicated.

### INTRODUCTION

A general theory of biological activity necessarily will be concerned with properties common to most biological materials. From the point of view of physics the extraordinary dielectric properties (1) are outstanding in this respect. Most remarkable amongst these are the very high electric fields of about  $10^7$  volt/m, usually present in biological membranes, leading to very strong polarization of the materials in them.

Biologically the term "life-cycle" already indicates a crude periodicity in biological dynamics, though here, each particular case exhibits individuality prohibiting the exact reproducibility of a period.

We shall take a vast step and assume that coherent excitation of electric vibrations in a large range of frequencies play an important role in biological activity. Modern physics offers this type of excitation arising through nonlinear interactions, with the necessary energy supplied, in general, from metabolic processes. Thus while supply of energy to a material usually leads to heating, coherent excitations may arise if certain conditions are satisfied.

This point of view was first presented more than fifteen years ago (2,3). It has led to far reaching consequences which stimulated a great number of experiments (review up to 1979 in (4)). In a recent meeting on coherent excitations in biological systems (5,6) the assembled physicists and biologists have arrived at the conclusion that coherent excitation had now been shown to be an intimate feature of biological activity\*.

#### COHERENT EXCITATIONS

Three types of coherent excitation have been proposed:

- A. Coherent excitation of a single polar mode.
- B. Excitation of a metastable highly polar state; this includes solitons.
- C. Vibrations arising from more complex processes and giving rise to limit cycles or Lotka-Volterra oscillations.

All three involve essentially nonlinear processes.

Frequencies of type A processes in membrane vibrations have been estimated in the  $10^{11}$  Hz region; this corresponds to the frequency of millimeter waves. Both, higher and lower frequencies of polar modes in biological molecules, and larger systems, do exist and must be relevant: Such vibrations interact with the

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\*Misrepresentation in this subject is fairly common. Thus in a recent book review in Bioelectromagnetics (December 1982) K.R. Foster seems to be quite unaware that experiments on the influence of millimeter waves on the rate of growth of yeast were undertaken to test the theoretical conjectures.

abundant cell water and thus will be strongly damped. Spectroscopy of non-excited systems thus shows broad bands only. Coherent excitation, however, as is well known from the theory of lasers, overcomes the damping and leads to sharp lines.

In the type A model, nonlinear interaction of the polar vibrations with a heat bath under the influence of energy supply at a rate  $s$  leads to the transformation of a Planck distribution of excitation at frequency  $s$  into a Bose distribution containing a chemical potential  $\mu$ , i.e. to a replacement of  $h\omega$  by  $h(\omega-\mu)$ . Here  $\mu$  increases with increasing  $s$  and at a critical value  $s_0$  causes Bose condensation, i.e. high excitation of a single mode\*.

A metastable highly polar state (B) may arise from the shape dependence of the electrostatic energy of a polarized material (7). Material deformation then must overcome the elastic energy, and the condition for the existence of the state B requires a certain softness which implies the sound velocity  $c$  to be below a certain critical value  $c_0$ . In extended regions where the sound velocity may vary spatially, those for which  $c < c_0$  may exhibit the excitation.

Excitation of type C involves the consequences of A and B and will be discussed below.

Excitation of a metastable state (B) requires a single appropriate energy supply. Excitation of coherent vibrations, however, as mentioned above involves a rate of energy supply  $s$  above a critical  $s_0$ . When the existing  $s$  is just below  $s_0$  then small further supply at the rate  $s_0 - s$  will induce the excitation.

It must also be pointed out that the frequency of the excited vibration need not be exactly reproducible. Assume an ion to be attached to the vibrating region in some cases but not in others. Through electrostatic interaction it may lead to a small shift in frequency through which in course of the nonlinear excitation may be stabilized. Also, nonlinear interaction may induce the frequency shifts which depend on the degree of excitation.

## CONSEQUENCES

Coherent excitation of a single polar mode (A) leads to frequency selective long range interaction between systems with

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\*Extensive theoretical work in this direction has been carried out by Bhaumik et al., Mills and Wu et al.; quoted in (4). Regrettably, very faulty theoretical work has also been published in this field.

equal frequencies; in biological membranes this will be in the  $10^{11}$  Hz region, but higher as well as lower frequencies, originating in other cell structures, will be expected. Such interaction may lead to selective attraction (or repulsion) of systems, to other control of their activities, and to communication within and between cells. It will be noted that electrostatic long range interaction is screened through counter ions.

Excitation of the metastable polar state (B) may lead to storage and transport of energy. It could represent the active state of enzymes and through the high internal field lead to a reduction of activation energies. It could also be effective in transporting energy from protein to protein. An array of proteins dissolved in a membrane would thus act as the energy transporting network postulated by current thinking (8).

The excitation discussed here may be switched on or off depending on the supply of energy at the appropriate time and place. In general this energy will arise from metabolic processes (or light in case of photosensitivity) and must be regulated by a control system. In the case of excitations A the energy must be supplied to the band of polar modes. According to the theoretical model, the system possesses storage capacity, and a certain time will be required to effect the excitation even if the rate of supply  $s$  exceeds the critical one,  $s > s_c$ . At this stage the system becomes sensitive to interference<sup>0</sup> from the outside. When the relevant band of frequencies is in the microwave, in particular in the millimeter wave region, then low intensity external radiation may cause large biological effects as the switching on of the excitation depends on  $s$  in a steplike fashion.

Such effects will be noticeable whenever the externally applied frequency falls into the band of polar modes. When a coherent mode begins to be excited, however, then energy application at its frequency will be more effective. Biological effects dependent on resonance must be expected. The external radiation then acts as a switch. It will be effective at certain periods of the biological development only.

The selective attraction arising in excitation A may play a crucial role in multienzyme processes. Assume enzymes to be spatially organized as discussed by Clegg in (6). Then some of the energy liberated during the enzymatic process may be used to cause excitation A in substrates which then will attract other substrates. This leads to an increase in available substrates and in turn to an increase in the rate of enzymatic processes which decreases the number of substrates. A typical Lotka-Volterra oscillation arises - characteristic of our type C of coherent

oscillations. More complex situations may arise leading to limit cycles as discussed in these proceedings by Kaiser. Further consequences are also discussed by Del Guidice.

Such excitations may also be sensitive to external vibrations which might, by relatively weak supply of energy, liberate the stored energy and thus cause large biological effects.

## EXPERIMENTS

Existence of three types of coherent excitations: A) a coherent single mode; B) a polar metastable state; C) a periodic activity arising from A and B, arises as the main conclusion of theoretical considerations. Detailed mechanisms might well differ considerably from the particular theoretical models; they might in fact be different in different cases.

Relevant experiments have been discussed in reference (6). They refer to the attraction of nutrients to a cell - similar to the attraction of a substrate as discussed above. The excitation is measured through non thermal excitation of anti-Stokes lines in Raman effect (Drissler et al. in (6). Rowland (6) gives evidence for the mutual attraction of blood cells satisfying the requirements of the theory.

Several contributions are related to biological effects arising from irradiation by very low intensity millimeter waves. Attempts to interpret these in conventional ways, e.g. in terms of the field turning a protein, are quite out of place as they would require (9) intensities of millions  $W/cm^2$  when experiments use 10  $mW/cm^2$  or less.

Experiments conducted so far do not permit specification of the particular biological process on which electromagnetic fields act. Although recently important evidence in this direction has been obtained, the experiments will have to be continued to establish this evidence. For example, extremely sharp frequency dependence of the influence of millimeter waves on the rate of growth of yeast have been reported by Grundler et al. (6). They find the step-like intensity dependence required by the theory. Apart from a very sharp resonance in the region of 41700 MHz, the frequency dependence exhibits a fine structure of about twenty minima and maxima, spaced at 7-8 MHz. It is tempting to interpret the basic line in terms of a type A excitation with the low frequency arising from processes of type C which require a type A excitation. With very sensitive instrumentation, carried out in an electrically screened laboratory, C.W. Smith and A.H. Jafary have measured very sharp emission in the 7-8 MHz region from single yeast cells. They occur during the period of cell division only, and last only about 3 minutes. The frequencies are not always

exactly the same, but this should be interpreted in terms of individuality as discussed in reference 2. We should conclude that the excitation is closely connected with the highly organized processes occurring in cell division.

Long range interaction may also be expected relevant for the control of cell division, important in the cancer problem (4) IVD). Experiments on growing tissue cultures would, therefore, be highly desirable. They do require, however, development of new experimental techniques.

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